

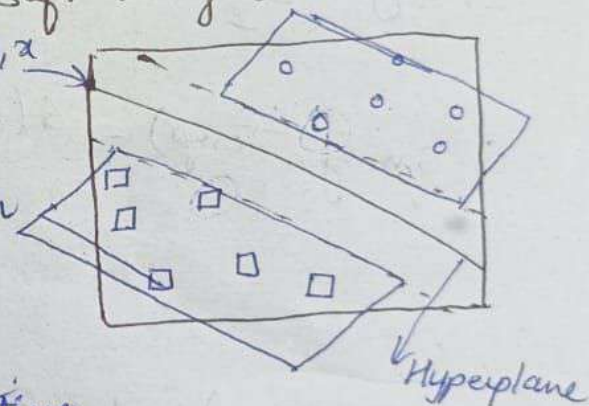
Support Vector Machines

- * Using concept of simple linear regression, support vector machines were designed.
- * Creating a boundary for classification
- * Decision Boundary
- * Hyperplane - Best divides the dataset into classes
 - line for 2D space, plane or a hyperplane for higher dimensions.
- * Margin - Dist b/w hyperplane & nearest points from either class
 - A larger margin implies a better generalization for the classifier.
- * Support vectors are the data points closest to the hyperplane

- 1) Hard Margin SVM → Without any error
- 2) Soft margin SVM → With some error

$$y = \beta_0 + \beta_1 x$$

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Adding a new dimension to support vectors → Augmentation

$$S_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad S_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad S_3 = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

Vector Augmentation

$$S_1 = (10) \quad \tilde{S}_1 = (101) \text{ Negative}$$

$$S_2 = (31) \quad \tilde{S}_2 = (311) \text{ Positive}$$

$$S_3 = (3-1) \quad \tilde{S}_3 = (3-11) \text{ Positive}$$

To segregate similar data → augmentation is used

$$WT = \beta_0 + \beta_1 x$$

$$\alpha_1 \tilde{S}_1 \tilde{S}_1 + \alpha_2 \tilde{S}_2 \tilde{S}_1 + \alpha_3 \tilde{S}_3 \tilde{S}_1 = -1$$

$$2\alpha_1 + 3 + 3 = -1 \Rightarrow \alpha_1 = -7/2$$

$$\alpha_1 \tilde{S}_1 \tilde{S}_2 + \alpha_2 \tilde{S}_2 \tilde{S}_2 + \alpha_3 \tilde{S}_3 \tilde{S}_2 = +1$$

$$4\alpha_1 + 8\alpha_2 + 8\alpha_3 = -2$$

$$\alpha_1 \tilde{S}_1 \tilde{S}_3 + \alpha_2 \tilde{S}_2 \tilde{S}_3 + \alpha_3 \tilde{S}_3 \tilde{S}_3 = +1$$

$$2\alpha_2 - 2\alpha_3 = 0 \Rightarrow \alpha_2 = \alpha_3$$

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1$$

$$3\alpha_2 + \alpha_3 = 3$$

$$4\alpha_1 + 9\alpha_2 + 9\alpha_3 = 1$$

$$\Rightarrow 4\alpha_2 = 3 \Rightarrow \alpha_2 = 3/4$$

support

$$\hat{W} = \sum_{i=1}^n a_i \tilde{s}_i$$

The last row will represent the bias.

$$\begin{aligned} \hat{W} &= -\frac{7}{2} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{3}{4} \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} + \frac{3}{4} \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \end{aligned}$$

$\begin{matrix} \nearrow & \rightarrow & \text{x-coefficient} \\ \rightarrow & \rightarrow & \text{y-coefficient} \\ \searrow & \rightarrow & \text{Bias} \end{matrix}$

Margin Calculus:-

Step 1:- Compute the Norm of (w)

$$\|W\| = \sqrt{1^2 + 0^2} = 1$$

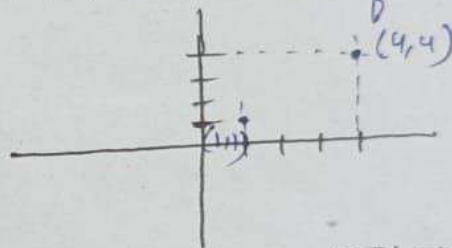
Step 2:- Compute the Margin

$$M = \frac{2}{\|W\|} = \frac{2}{1} = 2$$

Hyperplane is exactly in the middle of margin

Linear Kernels

4 4



* Whatever you are seeing may not be the right way for classification.

$$\begin{aligned} s_1 &= \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} & s_2 &= \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} \\ \tilde{s}_1 &= \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} & \tilde{s}_2 &= \begin{pmatrix} 2 & 2 & 0 \end{pmatrix} \end{aligned}$$

$$\alpha_1 \tilde{s}_1 \cdot \tilde{s}_1 + \alpha_2 \tilde{s}_2 \cdot \tilde{s}_1 = -1$$

$$\alpha_1 \tilde{s}_1 \cdot \tilde{s}_2 + \alpha_2 \tilde{s}_2 \cdot \tilde{s}_2 = 1$$

$$\Rightarrow 3\alpha_1 + \alpha_2 5 = -1$$

$$5\alpha_1 + 9\alpha_2 = 1$$

$$15\alpha_1 + 25\alpha_2 = -5$$

$$15\alpha_1 + 27\alpha_2 = 3$$

$$\Rightarrow 2\alpha_2 = 8 \Rightarrow \boxed{\alpha_2 = 4}$$

$$\alpha_1 = \frac{-1 - 5\alpha_2}{3} = \frac{-1 - 20}{3} = -7$$

$$\boxed{\alpha_1 = -7} \quad \boxed{\alpha_2 = 4}$$

$$-7 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}$$

ETL $\rightarrow$ Extraction, Transformation & Loading

$$x + y = 3 = 0$$

$$\|W\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$M =$$



$$= 0.55952$$

$$\text{Gini Index}(S, \text{Temp}) = 0.4591 - 0.55952$$

$$= -0.10042$$

$$G(S, \text{Humidity}) = 1 - \frac{7}{14} \left(1 - \left(\frac{3}{7} \right)^2 - \left(\frac{4}{7} \right)^2 \right) - \frac{7}{14} \left(1 - \left(\frac{5}{7} \right)^2 - \left(\frac{1}{7} \right)^2 \right)$$

$$= 1 - \frac{1}{2} \left(1 - \frac{25}{49} \right) - \frac{1}{2} \left(1 - \frac{37}{49} \right)$$

$$= 1 - \frac{1}{2} \left[\frac{24}{49} + \frac{12}{49} \right] = 1 - \frac{1}{2} \times \frac{36}{49}$$

$$= 0.63265$$

$$\text{Gini Index}(S, \text{Humidity}) = 0.4591 - 0.63265$$

$$= -0.17355$$

$$G(S, \text{Windy}) = 1 - \frac{8}{14} \left(1 - \left(\frac{6}{8} \right)^2 - \left(\frac{2}{8} \right)^2 \right) - \frac{6}{14} \left(1 - \left(\frac{3}{6} \right)^2 - \left(\frac{3}{6} \right)^2 \right)$$

$$= 1 - \frac{4}{7} \left(1 - \frac{10}{16} \right) - \frac{3}{7} \left(1 - \frac{1}{2} \right)$$

$$= 1 - \frac{4}{7} \times \frac{3}{8} - \frac{3}{7} \times \frac{1}{2}$$

$$= 1 - \frac{3}{14} - \frac{3}{14} = 1 - \frac{6}{14} = \frac{8}{14} = \frac{4}{7}$$

$$= 0.57143$$

$$\text{Gini Index}(S, \text{Windy}) = 0.4591 - 0.57143$$

$$= -0.11233$$

Min Gini Index $\rightarrow$ Root Node

Temperature

	Temp	Humidity	Windy	Play Golf
1	Hot	High	Weak	N
2	Hot	High	Strong	N
3	Mild	High	Weak	Y
4	Cool	Normal	Weak	Y
5	Mild	Normal	Strong	X

$$1 - \left(\frac{6}{7}\right)^2 - \left(\frac{1}{7}\right)^2$$

Temp		yes	No	P(yes)	P(No)
Rain	Hot	0	2	0	$\frac{2}{3}$
	Mild	1	1	$\frac{1}{2}$	$\frac{1}{3}$
	Cool	$\frac{1}{2}$	$\frac{0}{3}$	$\frac{1}{2}$	0
				100%	100%

Humidity		yes	No	P(yes)	P(No)
High		0	3	0	1
Normal		$\frac{2}{2}$	$\frac{0}{3}$	1	0

Windy		yes	No	P(yes)	P(No)
Weak		1	2	$\frac{1}{2}$	$\frac{2}{3}$
Strong		$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$
				100%	100%

$$G(\text{Sunny}, \text{Temp}) = G(\text{Rainy}) = 1 - \left(\frac{3}{5}\right)^2 - \left(\frac{2}{5}\right)^2$$

$$= 1 - \frac{9}{25} - \frac{4}{25} = 1 - \frac{13}{25}$$

$$= \frac{12}{25}$$

$$= 0.48$$

$$G(\text{Rainy}, \text{Temp}) = 1 - \frac{2}{5} \left(1 - \left(\frac{2}{3}\right)^2\right) - \frac{2}{5} \left(1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2\right)$$

$$- \frac{1}{5} \left(1 - \left(\frac{1}{2}\right)^2\right)$$

$$= 1 - \frac{2}{5} \times \frac{5}{9} - \frac{2}{5} \left(1 - \frac{13}{36}\right) - \frac{1}{5} \left(1 - \frac{3}{4}\right)$$

$$= 1 - \frac{2}{9} - \frac{2}{5} \times \frac{23}{36} - \frac{3}{20} = 0.3722$$

$$\text{Grind/Ingen } G(\text{Rainy}, \text{Humidity}) = 1 - \frac{3}{5} \left(1 - (1)^2\right) - \frac{2}{5} \left(1 - (1)^2\right)$$

$$= 1$$

$$G(\text{Rainy}, \text{Windy}) = 1 - \frac{3}{5} \left(1 - \left(\frac{1}{2}\right)^2 - \left(\frac{2}{3}\right)^2\right) - \frac{2}{5} \left(1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2\right)$$

$$= 1 - \frac{3}{5} \left(1 - \frac{25}{36}\right) - \frac{2}{5} \times \frac{11}{12} = 1 - \frac{8}{12} - \frac{11}{30} = \frac{1}{5}$$

$$\frac{1 \times 9}{4 \times 9} + \frac{4 \times 4}{9 \times 9}$$

$$E' (A_0, B_1)$$

$$0.8113$$

852
5917

py(x)

$$\text{Gini Impurity} = 1 - \sum_{i=1}^n (p_i)^2$$

$$\text{Gini Impurity} = 1 - \sum_{i=1}^n (p_i) * (1 - p_i)$$

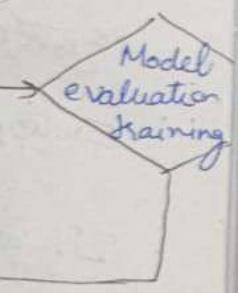
Gini Index = Gini(S) - Gini Impurity (S, X)

- * In Gini Index, height might be more.
- * Calculation is more in case of IG.

$$G(S) = 1 - \left[\left(\frac{9}{14} \right)^2 + \left(\frac{5}{14} \right)^2 \right] = 1 - \left(\frac{81+25}{196} \right)$$

$$= 1 - \frac{106}{196} = \frac{90}{196} = 0.4591$$

$$G(S, \text{outlook}) = 1 - \left[\frac{5}{14} G(\text{outlook}, \text{Sunny}) + \frac{4}{14} G(\text{outlook}, \text{Overcast}) + \frac{5}{14} G(\text{outlook}, \text{Rainy}) \right]$$



$$= 1 - \frac{5}{14} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right) - \frac{4}{14} \left(1 - \left(\frac{4}{4} \right)^2 - \left(\frac{0}{4} \right)^2 \right) - \frac{5}{14} \left(1 - \left(\frac{2}{5} \right)^2 - \left(\frac{3}{5} \right)^2 \right)$$

$$= 0.657143$$

	Overcast		Count
	Yes	No	
Sunny	3	2	5
Overcast	4	0	4
Rainy	2	3	5
Total	9	5	100%

$$G(S, \text{Gini Index } (S, \text{outlook})) = 0.4591 - 0.657143 = -0.19804$$

$$G(S, \text{Temp}) = 1 - \frac{4}{14} \left(1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right) - \frac{6}{14} \left(1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right) - \frac{4}{14} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right)$$

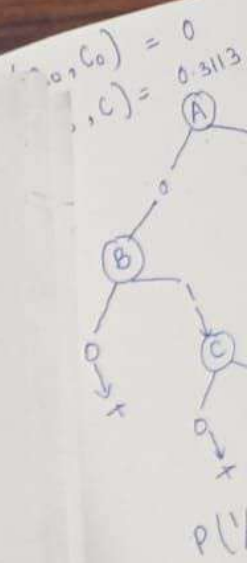
$$\frac{4}{9} + \frac{1}{9}$$

$$= 1 - \frac{4}{14} \left(1 - \frac{1}{2} \right) - \frac{6}{14} \left(1 - \frac{5}{9} \right) - \frac{4}{14} \left(1 - \frac{10}{16} \right)$$

$$= 1 - \frac{2}{7} \times \frac{1}{2} - \frac{3}{7} \times \frac{4}{9} - \frac{2}{7} \times \frac{3}{8} = 1 - \frac{1}{7} - \frac{4}{21} - \frac{3}{28}$$

$$E(A_0, B) = 0.811$$

$$E(A_0, B) = 0.3113$$



5113 No No = 1